

EPJ B

Condensed Matter
and Complex Systems

EPJ.org

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Eur. Phys. J. B (2012) 85: 234

DOI: 10.1140/epjb/e2012-30069-3

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Received 28 January 2012 / Received in final form 16 April 2012

Published online 3 July 2012 – © EDP Sciences, Società Italiana di Fisica, Springer-Verlag 2012

Abstract. The quantum paraelectric strontium titanate can be made ferroelectric through replacing oxygen atoms with their heavy isotopes. Although suppressed quantum fluctuations have been widely believed to be the origin, the details remain unsettled. The controversies are often framed using Barrett formula, which involves two quantities termed T_0 and T_1 , respectively. The observations can be equally explained by assuming either a decrease in T_1 or an increase in T_0 upon isotope replacement. The conventional view holds a direct connection between quantum fluctuations and the T_1 and hence adopts the decreasing T_1 picture. In this paper, we offer a different opinion, in which the T_1 bears a different meaning and quantum fluctuations are attached to another quantity to be denoted ω_0 . We show that a decrease in ω_0 could diminish quantum fluctuations and simultaneously enhance T_0 . A vibronic mechanism is presented as a possible route to the ω_0 change. The isotope effects are argued to be rather non-local. The dynamics of the system could be highly quasi-one-dimensional and this is then employed in discussing the relation between a soft mode and a recently observed central mode.

Strontium titanate (STO) may be among the most extensively studied members of the oxide perovskite family. Its static dielectric constant ϵ was found to obey the Barrett formula [1],

$$\epsilon = \frac{C}{\frac{T_1}{2} \coth\left(\frac{T_1}{2T}\right) - T_0} \quad (1)$$

where $T_{1,0}$ denote temperature parameters and C the Curie constant. This is often taken as an evidence for strong quantum fluctuations (QF), which are usually supposed to be measured by the T_1 . QF set in at low temperatures and hinder the ferroelectric instability that is already seen at high temperatures. According to this logic, one expects ferroelectricity (FE) to take place if the QF can be suppressed.

In 1999, Itoh et al. [2] found that FE could be induced in STO by isotope substitution: replacing O^{16} with O^{18} by a fraction in excess of about $x_c = 33\%$ leads to an FE transition. The compound after substitution may be formulated as STO_x for short, with x being the fraction of O^{18} . The corresponding transition temperature T_c as a function of x fits that

$$T_c(x) = 30.4 \text{ K}(x - x_c)^{1/2}, \quad (2)$$

which yields $T_c \approx 24 \text{ K}$ for STO_{18} (i.e., the compound at $x = 1$). As for the origin of this isotope exchange induced FE, suppression of QF has been widely quoted. However, it remains elusive how such suppression occurs.

In another work [3], Itoh et al. argued that the temperature dependence of the static dielectric constant of STO_x could be reproduced if the so-called critical quantum displacive limit had been realized. To fit Barrett formula, they kept T_0 constant while varying only T_1 in the range from 71 K to 87 K as x changes. However, this scheme is questionable. According to reference [4], in the quantum displacive limit the T_1 is actually proportional to a frequency called ω_{loc} (in Ref. [4]), which also enters T_0 . Therefore, both T_1 and T_0 have to change with x under this scheme. Moreover, as reasoned in reference [5], it is unrealistic to have a T_1 change as large as 25% while the mass change of TiO_3 is less than 6%.

Bussmann-Holder and Bishop [6] with their non-linear shell model [7,8] invoked an intrinsic inhomogeneity to explain the huge isotope effect. Their model captures essentially the Slater mode but with additional anharmonicity. To achieve the observed T_c , the mass change needed has to be as large as 40%, as commented by Yacoby and Girshberg [5]. Yamada et al. proposed a three-state model based on a bi-polaron picture (whose existence certainly requires further clarifications) for a TiO_6 cluster, which is assumed to move in a triple-well potential that has a deep main well sandwiched by two shallow side wells [4]. A tunneling is imposed to connect the side wells. They further assume that the isotope exchange somehow decreases the tunneling frequency and therefore induces FE in their scheme. Note that this frequency has to change as much as 15%, nearly four times the mass change of TiO_6 , to meet the need.

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Yacoby and Girshberg [5] suggested an explanation within their spin-phonon jargon [9–12]. In their model, the Ti atoms are described by spin variables while simultaneously a soft phonon component is employed to interact with the spins. Judged from the relations these authors adopted to evaluate the phonon frequency [5], one may allege that the phonon component in their model should also be related to the Slater mode. The frequency of this soft phonon is expected to decrease as O is replaced with its heavy isotope and this decrease is then expected to enhance the effective spin-spin coupling. As a result, FE may be eventually induced. In this interpretation, the QF have been left intact from isotope substitution. Despite its success, there seems a loophole with their argument. Although isotope substitution decreases the phonon frequency, at the same time the related mass must increase by exactly such an amount that the force constant, which is proportional to the mass multiplied by the frequency squared, remains unchanged. The effective spin-spin coupling is expected to be proportional to $\frac{1}{\text{mass} \times \text{frequency}^2}$, rather than to $\frac{1}{\text{frequency}^2}$. Therefore, the scheme as desired by Yacoby et al. could not work.

In what follows, we present a mechanism allowing for the observed phenomena. It is based on the picture we forwarded in a previous publication [13], where we constructed an explicit atomistic model including both Ti- and O-sublattice. The exclusion of the Sr-sublattice was justified there utilizing the partial density of phonon modes. The model puts strong anharmonicity on the Ti atoms while assuming perfect harmonicity for the O atoms, consistent with observations. The Ti-O interactions were assumed to exist between nearest Ti and O atoms and take a simple bilinear form in displacements. By virtue of symmetry and other reasoning [13], the O vibrations were found anisotropic in the sense that, the frequency for vibrations along Ti-O-Ti bonds differs from that in the perpendicular directions. With this model, the observed dielectric modes in STO can be deciphered directly [13]. In the high temperature cubic phase, three modes have been experimentally known for long [14]. The two with frequencies at 178 cm^{-1} (TO_2) and 546 cm^{-1} (TO_4), respectively, can be attached to the O vibrations along and perpendicular to the Ti-O-Ti bonds, respectively [13]. The lowest mode (TO_1) can then be matched with Ti motions [13]. As temperature goes down, this mode was observed to soften quickly as anticipated due to anharmonicity, whereas the TO_2 and TO_4 were seen nearly unchanged [15]. Across $T_a = 105 \text{ K}$, TO_1 splits into a singlet (TO_A , pertaining to c -axis) and a doublet (TO_E to a - b plane) due to the tetragonal distortion accompanying the anti-ferro-distortive (AFD) transition that occurs at T_a involving the O octahedra rotation [16,17]. The TO_A lies above TO_E in frequency, with the discrepancy growing under further cooling [16,17]. Therefore, TO_E should be the mode that is directly associated with the ferroelectric instability under interest, as demonstrated in recent studies [18–20].

Despite demonstrated influences of the AFD transition upon the ferroelectric instability [16,17,21,22], we do not see immediate need to consider these influences *explicitly*, which seem mainly to cause the mode splitting and to prevail an FE with spontaneous polarization in the a - b plane. It is not unreasonable to imagine that, an FE could occur with polarization in any axial direction with the same likelihood if not for the AFD transition, as clearly indicated in the work by Yamanaka et al. [17]. This means that, the ferroelectric instability, albeit affected by the AFD transition, should be fundamentally and essentially captured by the soft phonon associated with the anharmonic motions of Ti atoms in any axial direction. The splitting can be either ignored or taken care of by using an averaged frequency (to be denoted by ω_s) for the soft mode [16]. Therefore, we proceed to examine the motions along e.g. z -axis.

There is some controversy over the modeling of Ti motions. Vogt stated that a Ti atom should be subjected to a single-well potential [16]. On the other hand, NMR experiments [23] seem to suggest a multi-well potential, which has promoted Yacoby et al. to model a Ti atom as a spin in a virtual transverse magnetic field. However, the well separation and hence the well depth should be very small [24] and no intra-well motions can be defined (which agrees with the fact that only three dielectric peaks can be observed in STO even at very low temperatures). Under such circumstances, one may approximate the multi-well potential by an effective single-well potential and a Ti atom can then be taken as a simple harmonic oscillator whose frequency ω_s is expected to depend on temperature. This dependence can often be evaluated by e.g. self-consistent harmonic approximation [16,25]. As implied in these work, one may write

$$\omega_s^2 = \omega_0^2 \coth \frac{T_1}{2T} \quad (3)$$

where ω_0 is the zero temperature frequency that represents quantum tunneling and T_1 is a parameter that basically accounts for the nature of the anharmonicity and will enter the Barrett formula. As to be seen, the ω_0 for STO (hereafter denoted by Ω) will be different from that for STO18 (by Ω^*).

The total energy of the TiO_3 subsystem is then written in terms of the z -components of the Ti and O displacements and their derivatives as

$$E = \frac{M}{2} \sum_i (\dot{Z}_i^2 + \omega_s^2 Z_i^2) + \frac{m}{2} \sum_{i,\rho} (\dot{z}_{i\rho}^2 + \omega_\rho^2 z_{i\rho}^2) + \sum_{i,j,\rho} \beta_{\rho;ij} Z_i z_{j\rho}. \quad (4)$$

Here M and m stand for the mass of Ti and O atoms, respectively, the upper dot denotes the time derivative, the $\rho = x, y, z$ label the three inequivalent O positions in a unit cell, ω_ρ give the frequencies of O vibrations and $\beta_{\rho;ij}$ signifies the coupling between the i th Ti atom and the j th O atom. As aforementioned, we have $\omega_z = \omega_l$ and $\omega_x = \omega_y = \omega_t$, with ω_l and ω_t denoting the O vibration frequencies along and perpendicular to Ti-O-Ti

bonds, respectively. Besides, for an adjacent pair of (Ti,O) lying along z -axis we choose $\beta_{z;ij} = \beta_l$ while in the x - and y -axis $\beta_{x,y;ij} = \beta_t$ and otherwise this quantity vanishes.

We are interested in the long wavelength dynamics concerned with the zone-center mode, for which one easily obtains the following dynamical matrix as a function of frequency ω ,

$$\Xi(\omega) = \begin{pmatrix} G_s^{-1}(\omega) & 2\beta_l & 2\beta_t \\ 2\beta_l & G_l^{-1}(\omega) & 0 \\ 2\beta_t & 0 & G_t^{-1}(\omega) \end{pmatrix} \quad (5)$$

where $G_s(\omega) = \frac{1}{M(\omega_s^2 - \omega^2 - i\omega\Gamma_s)}$ and $G_{l,t}(\omega) = \frac{1}{m(\omega_{l,t}^2 - \omega^2 - i\omega\Gamma_{l,t})}$ may be called the Green functions of the s - and l - and t -mode, respectively. To account for dissipation effects, we have introduced these damping constants Γ and Γ_ρ . In terms of $\Xi(\omega)$, the dielectric function can be obtained as

$$\epsilon(\omega) = \epsilon_\infty + \frac{e^2}{\epsilon_0 a_0^3} Q' \Xi(\omega) Q \quad (6)$$

where $\epsilon_\infty \approx 5.2$ is negligible in comparison with the second contribution, e is the elementary charge and a_0 the lattice constant while ϵ_0 is the vacuum permittivity. In addition, Q is the charge column in units of e and $Q' = [Q_s, Q_l, Q_t]$ denotes the transpose of Q . The static dielectric constant follows at $\omega = 0$ and it takes on the form of equation (1), with the Curie constant given by

$$C = \frac{e^2 l_1^2}{k_B a_0^3 \epsilon_0} \left\{ Q_s^2 - 8 \frac{(Q_l \beta_t + Q_t \beta_l)^2}{m^2 \omega_l^2 \omega_t^2} + \frac{\frac{Q_l^2 k_B T_1 \coth \frac{T_1}{2T}}{2l_1^2} - 4Q_l Q_s \beta_l}{m \omega_l^2} + 2 \frac{\frac{Q_t^2 k_B T_1 \coth \frac{T_1}{2T}}{2l_1^2} - 4Q_t Q_s \beta_t}{m \omega_t^2} \right\} \quad (7)$$

and the extrapolated Curie temperature as

$$T_0 = 4k_B^{-1} l_1^2 \beta, \quad \beta = \frac{2\beta_t^2}{m \omega_t^2} + \frac{\beta_l^2}{m \omega_l^2}. \quad (8)$$

Now the transition temperature T_c is determined by the divergence of ϵ , namely,

$$\frac{T_1}{2} \coth \frac{T_1}{2T_c} = T_0. \quad (9)$$

which has no solution unless $T_1 < 2T_0$.

Note that in the above expressions there involves an interesting length scale l_1 , which reads

$$l_1 = \sqrt{\frac{k_B T_1}{2M \omega_0^2}}. \quad (10)$$

This is the only quantity where ω_0 appears directly. This length scale is intrinsic to the Ti atoms, since it is expressed sheerly in terms of quantities characterizing these

atoms. See that ω_0 is the value of ω_s at vanishing temperature, which therefore should be related to the zero-point QF. A larger ω_0 indicates stronger QF. In the meanwhile, T_1 is another independent parameter having nothing to do with ω_0 , contrary to the conventional view that these two quantities be proportional, and it obviously characterizes the strength and category of relevant anharmonicity. l_1 can roughly be regarded as the length by which a Ti is displaced at zero temperature to have an elastic energy of $k_B T_1/4$.

We proceed to find out all the parameters. The lattice constant was experimentally known to be around $a_0 \approx 400$ pm. As for the charges, first-principles computations give $Q_s = 7.25$, $Q_l = -5.7$ and $Q_t = -2.15$ [26]. The work by Itoh et al. give that $T_1 = 84$ K for STO, which in combination with the value of $\omega_s \approx 87.7$ cm⁻¹ at room temperature [14] yields via equation (3) that $\Omega \approx 32.7$ cm⁻¹ for STO. Hence, $l_1 \approx 13.8$ pm for STO. Substituting these in equation (7), we arrive at

$$\frac{C}{0.0062 \times 10^5 K} \approx 56.0227 + 5.5317\beta_l + 0.4412\beta_t. \quad (11)$$

Moreover, as previously discussed, ω_l and ω_t should correspond to the frequencies of TO₂ and TO₄, respectively. Therefore, $\omega_l \approx 178$ cm⁻¹ and $\omega_t \approx 546$ cm⁻¹. With $T_0 \approx 38$ K quoted from reference [2] and M being the Ti mass, equation (8) can be translated to be

$$0.6878 \approx 0.0071\beta_t^2 + 0.0332\beta_l^2. \quad (12)$$

These two equations (11) and (12) can be gratified if we choose $\beta_l \approx 4.4 \times 10^3$ g s⁻² and $\beta_t \approx 2.5 \times 10^3$ g s⁻², which gives $C \approx 0.6 \times 10^5$ K, comparable to the experimental value [1].

Now we discuss the O isotope effects within the above context. Relevant to equation (9) are T_0 and T_1 . How should they change under isotope replacement? In their early report [2], Itoh et al. varied both to fit the data of STO18, for which they took $T_1^* = 87$ K and $T_0^* = 44$ K (here the star indicates the reference to STO18). However, later they varied only T_1 to match the observations [3] and concluded that increasing x decreases QF. In their reasoning, T_1 was obviously taken for granted to represent the zero point fluctuations. Nevertheless, according to the present scheme, these fluctuations should directly pertain to ω_0 . As x increases, we then expect a reduction in ω_0 . Adhering to this scheme, we from now on tentatively put T_1 constant, i.e., $T_1^* = T_1 = 84$ K. The FE transition in STO18 was known to occur about 24 K [2]. From equation (9), one finds $T_0^* \approx 44.62$ K, very close to the value reported in reference [2]. Comparing STO with STO18, we are led to

$$\frac{2T_0}{T_1} \left(\frac{2T_0^*}{T_1^*} \right)^{-1} = \frac{T_0}{T_0^*} = \frac{\beta}{\beta^*} \frac{\Omega^{*2}}{\Omega^2} \quad (13)$$

where we have used equations (8) and (10). Reasonably, the isotope exchange should affect neither $\beta_{l,t}$ nor $m \omega_{l,t}^2$, since these quantities are force constants that are primarily determined by the crystal and the electronic

structure. Hence, we should have $\beta = \beta^*$, which yields $\Omega^* = \Omega \sqrt{\frac{T_0}{T_0^*}} \approx 30.18 \text{ cm}^{-1}$. Relative to Ω , Ω^* has therefore been diminished by about 7.7%.

The above analysis shows that QF can be weakened by reducing the value of ω_0 , a quantity that measures the zero point energy carried by Ti motions at vanishing temperature. This reduction leads to an increase in l_1 , which in turn enhances the effective molecular field strength as represented by T_0 to such an amount that equation (9) admits an FE solution. At the same time the T_1 can be held invariant. The question is, how could isotope replacement result in a smaller ω_0 ? Naively, one may attribute the ω_0 reduction to an enhancement in the effective mass associated with Ti motions. Nonetheless, this scenario is unlikely to bring about a change in l_1 (and T_0): what enters in l_1 is the force constant $M\omega_0^2$ (not just ω_0^2), which, as we have emphasized in the above, is an invariant under isotope exchange.

In another scenario, the ω_0 could be altered by changing the Ti potential surface. Recent studies [16,17] demonstrated that a small rotation of the O octahedra (as involved in the AFD transition) could significantly impact the Ti behaviors and especially, it might modify the Ti potential surface. In reference [22], a Ti atom was modelled as a rotor moving on a non-uniform sphere and the O octahedra rotation was shown to influence the property of this sphere. Here we prove from a vibronic point of view that intimate relations indeed exist between the O octahedra and the Ti potential surface. For illustration, it suffices to consider a single TiO_6 cluster. As shown by Bersuker [27–30], the Ti atom in this cluster tends to be off-center shifted so as to lower the electronic energy of the system (the so-called Pseudo-Jahn-Teller effect). He noted that, the off-center shift could break certain reflection symmetries and therefore allow electronic transitions to occur between the Ti- t_{2g} orbital (i.e., d_{xy} , d_{yz} and d_{zx}) and the O- σ -type p orbital (i.e., $p_{+;x,y,z}$ and $p_{-;x,y,z}$, with $\pm$ indicating the O sites on the same line); these transitions then effectively lower the p -orbital energy. Interestingly, we observe that, the same transitions can be caused by tilting the O octahedra, as such tilting breaks the same symmetry. Suppose the octahedra is rotated about z -axis by a small angle φ and simultaneously the Ti atom is slightly displaced by $\vec{u} = (X, Y, Z)$. Within the Born-Oppenheimer approximation, the effective potential energy for this configuration can be obtained by solving the following model Hamiltonian:

$$\begin{aligned}
 H = & \frac{\Delta}{2} \left(\sum_{i=xy,yz,zx} d_i^\dagger d_i - \sum_{i=x,y,z} P_i^\dagger P_i \right) \\
 & + \left\{ d_{xy}^\dagger \left[P_x \left(\xi Y + \varsigma \frac{a_0}{2} \varphi \right) + P_y \left(\xi X + \varsigma \frac{a_0}{2} \varphi \right) \right] \right. \\
 & + h.c. \left. \right\} + \left\{ d_{yz}^\dagger \xi (P_y Z + P_z Y) + h.c. \right\} \\
 & + \left\{ d_{zx}^\dagger \xi (P_z X + P_x Z) + h.c. \right\} \quad (14)
 \end{aligned}$$

where $d_i^\dagger$ and $P_i^\dagger$ (d_i and $P_i \equiv \frac{1}{\sqrt{2}}(p_{+;i} - p_{-;i})$) generates (annihilates) an electron on the corresponding orbital i ,

respectively, and ξ together with ς specifies the vibronic coupling strengths. Besides, Δ signifies the energy spacing between the d - and p -orbital and $h.c.$ stands for the Hermitian conjugate.

Obviously, this H reduces to the Bersuker model when $\varphi = 0$ [27]. Its eigenstates can be grouped into three bonding states with energies $-\varepsilon_{1,2,3}$ and three anti-bonding states with energies $\varepsilon_{1,2,3}$. The effective potential energy is then given by

$$\varepsilon_{eff}(\vec{u}, \varphi) = K_{\text{Ti}} \vec{u}^2 + K_O a_0^2 \varphi^2 - 2 \sum_{i=1,2,3} \varepsilon_i. \quad (15)$$

Here the first two terms are included to represent the elastic energy and the factor 2 takes care of the spin variables. Let $\hat{\varepsilon}_i = \frac{\varepsilon_i^2 - \Delta^2}{\xi^2}$ and $L = \frac{a_0 \varsigma \varphi}{2\xi}$. Through block matrix method one can show that the secular equation corresponding to H reads

$$\begin{aligned}
 \hat{\varepsilon}^3 - 2\vec{u}^2 \hat{\varepsilon}^2 + \vec{u}^4 \hat{\varepsilon} + 2L(X + Y + L)\hat{\varepsilon}(\vec{u}^2 - \hat{\varepsilon}) = \\
 4Z^2 X^2 Y^2 + 4Z^2 XY L(X + Y) + Z^2 L^2(X + Y)^2 \quad (16)
 \end{aligned}$$

which has three positive roots, as required. Explicit expressions of $\hat{\varepsilon}_i$ are not needed for the present purpose. As we notice, in STO the AFD transition occurs around 105 K whereas in STO18, it occurs about 107 K, slightly higher than in STO. In either compound, the AFD transition temperature is much higher than the FE transition temperature, implying that the AFD and FE transitions are well decoupled in energy. Thus, the Ti potential surface can be simply obtained by averaging over the φ , i.e., $\varepsilon_{Ti}(\vec{u}) = \langle \varepsilon_{eff}(\vec{u}, \varphi) \rangle_\varphi$. Below the AFD transition temperature, this averaging amounts to fixing φ at its thermodynamic average. Hence, $\varepsilon_{Ti}(\vec{u}) \approx \varepsilon_{eff}(\vec{u}, \langle \varphi \rangle)$. Now that it affects $\langle \varphi \rangle$, the O isotope exchange is bound to affect the Ti potential surface ε_{Ti} .

We go on to investigate the x dependence of T_c . To this end, one has to know how to choose a proper representative of ω_0 , which can be written as $\bar{\omega}$. At $x = 0$, the ω_0 can only be Ω while at $x = 1$, it can only be Ω^* . In such extremes, the compound has full translational symmetry. However, for a general x in between 0 and unity, this symmetry is broken and the value of ω_0 must vary randomly from site to site in the Ti-sublattice. Moreover, it is not guaranteed that given a site, the ω_0 should be either Ω or Ω^* . Therefore, there is no simple way to establish a formula for $\bar{\omega}$. To make progress, however, we assume that it can be obtained by a proper average. To this end, we introduce a function of x , $p(x)$, which measures the likelihood of a randomly picked Ti atom whose potential surface resembles that in STO18. In this paper, the average is taken over $\frac{1}{\omega_0^2}$, in stead of e.g. ω_0^2 . It turns out that, the details of such choice do not affect the basic physics under interest. Thus, we have

$$\frac{1}{\bar{\omega}^2} = p(x) \frac{1}{\Omega^{*2}} + [1 - p(x)] \frac{1}{\Omega^2}. \quad (17)$$

By definition, one has $p(0) = 0$ and $p(1) = 1$.

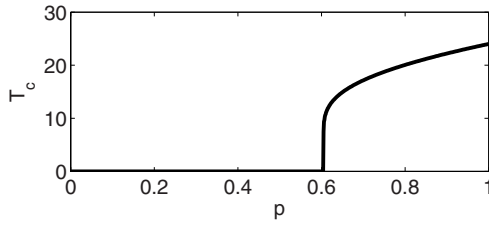


Fig. 1. Transition temperature T_c (K) as a function of p .

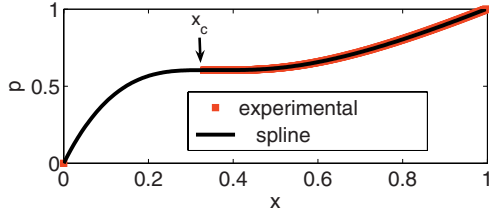


Fig. 2. (Color online) The inferred $p(x)$. The squares are explained in the text, and the solid line is a simple spline fitting.

Now substituting $\bar{\omega}$ for ω_0 in equations (8)–(10), we find T_c for every p . The T_c - p curve is plotted in Figure 1, where it is seen that the critical probability occurs at $p_c \approx 0.6$. So, p_c is as twice large as x_c , suggesting an initial rapid increase in p as x varies. To confirm this, we have inferred the function $p(x)$, as exhibited in Figure 2. This is achieved by numerically solving equation (9) to obtain $\bar{\omega}$ for every x , in which the T_c is evaluated via equation (2). Then the $\bar{\omega}$ is utilized to deduce p (the red squares in Fig. 2). The as-found $p(x)$ features a turning section where x_c belongs. Below x_c , p increases very fast while above that point it increases only slowly. Implications seem to be that, the exchange effects are non-local: an O replacement around one Ti atom affects not only this atom but also distant ones. A few heavy O atoms seem capable of setting up a network that influences a number of Ti atoms.

Finally, we discuss another issue relevant to the induced FE transition. Shigenari et al. noticed a central mode they held directly responsible for the transition [18]. Beside this mode, they also found another mode which softens but not perfectly: it is pinned at a low frequency at temperatures close to T_c . The central mode was shown of relaxation type and has been ascribed to polar cluster dynamics [18]. We try to gain some insights into such phenomena within the present understanding. It is seen that the β in equation (8) consists of two contributions: $K_l \equiv \frac{\beta_l^2}{m\omega_l^2}$ and $2K_t \equiv 2\frac{\beta_t^2}{m\omega_t^2}$. The factor 2 comes from the degeneracy within the xy -plane. Actually, K_l and K_t represent the effective inter-Ti couplings (mediated by the shared O atom) along z - and $x(y)$ -direction, respectively [13].

With the parameters prescribed above, we find a strong anisotropy in the Ti-Ti couplings: $\eta = \frac{K_l}{K_t} \approx 29$. As a result, robust quasi-one-dimensional correlations along [001] directions are anticipated [31]. These correlations then lead to chain-like structures ($-\text{[Ti-O]}_n-$, n being the average chain size) at temperatures even far above T_c . Notably, such structures have been known for decades in the prototype ferroelectric barium titanate and they were found crucial in apprehending some key features (such as

the small entropy change) of the cubic-tetragonal transition in this compound [32,33]. A crude estimate of n can be obtained by mean-field reasoning. On one hand, the mean-field strength about a Ti atom is $\sim 2K_l + 4K_t$, since it has four neighbors in the xy -plane and two neighbors along z -axis. On the other hand, the mean-field strength about a chain is $\sim 4nK_t$. These two strengths have to be equal, i.e., $2K_l + 4K_t \sim 4nK_t$, whence $n \sim 1 + \frac{\eta}{2} = 16$. Therefore, a chain covers about 16 unit cells for temperatures near T_c . A more systematic estimate can be made with the method used in reference [31].

In the paraelectric phase, the chains are randomly oriented. Within each chain the Ti atoms are strongly (as characterized by K_l) correlated in their displacements along the chain and therefore one-dimensional lattice vibrations must be present. Upon cooling, the lowest frequency vibrations get soft so that ordering could occur within a chain. These vibrations are carried with Ti atoms and hence a proper description may be obtained by elimination of O motions. Through adiabatic decoupling [13], an effective model can be established for intra-chain Ti motions: $\frac{M}{2} \sum_i (\dot{Z}_i^2 + \omega_s^2 Z_i^2) - K_l \sum_{\langle i,j \rangle} Z_i Z_j$, where the i runs over an entire chain. The normal modes of this model can be labelled by wavenumbers $q = \frac{r\pi}{n}$, with $r = 0, \pm 1, \pm 2, \dots$. Their frequencies are given by $\omega(q)^2 = \omega_s^2 - 2K_l \cos(q)$. The lowest mode occurs at $q = 0$, with frequency $\omega^2(q = 0) = \omega_s^2 - 2K_l$, which obviously softens as temperature goes down but never reaches zero above T_c .

At the same time, there exists inter-chain correlations, which are ultimately responsible for inter-chain ordering. This ordering can take place only in a very incoherent way and thus must give rise to a relaxation mode. As a simple modelling, we may view each chain in its ground state as a bistable object: its polarization can point in two directions along the chain. Thus, the inter-chain interactions can be formalized as $-nK_t b^2 \sum_{\langle i,j \rangle} \sigma_i^z \sigma_j^z$, where i and j label the chains and $\sigma^z = \pm 1$. In addition, we have introduced a parameter b to measure the average Ti displacement of a chain. We see that this model does not admit intrinsic dynamics. Therefore, one has to couple the system to a heat bath to have dynamical phenomena (e.g. to drive the system toward thermodynamic equilibrium), which can only be totally incoherent. Such coupling is often considered by a phenomenological relaxation time τ_0 , which describes roughly the duration of an isolated Ti atom in one of its possible states, say $\sigma^z = 1$. According to Glauber [34], one expects in the dynamical susceptibility function a relaxation mode with a frequency given as $\tau_0^{-1} (1 - \frac{4nK_t b^2}{k_B T})$, which shifts to zero as T approaches T_c .

It is pointed out that, the coexistence of a resonance mode and a relaxation mode in a strongly anisotropic system has actually been well understood in reference [31,35], where a quasi-1D Ising model was investigated. Experiments are highly desirable to test this picture.

In summary, the model proposed above has been shown probable in offering a consistent explanation of the most fundamental aspects of the isotope exchange induced FE in STO.

This work was supported by grants from the Hong Kong Polytechnic University (Project Nos. A-PK26 and RPQ1).

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